陸唯佳

Weijia Lu is a chief researcher with two PhDs and a senior manager of an innoventional team of diverse technical directions;
He has over 20 years of experiences in healthcare and automotive;
And demonstrated excellence in AI, multiphysics numerical analysis, signal processing and likely.

CONTACT

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SKILLS

Industry
Healthcare
Automotive

Leadership & Management
Strategic Planning
Quality Assurance
Team Leadership
Visionary Thinking

Research & Delivery
Signal Processing
(e.g. Image, Medical Signal, Text ...)
Deep/Machine Learning
(e.g. GAN, RL, Plant Modeling ...)
Compute Architecture
(e.g. Model Pruning, FL ...)
Physics & Modeling
(e.g. FEM, FIELD II, Abersim, COMSOL ...)
Software Implementation
(e.g. Python, C/C++, R, Matlab, GNU Tools, Linux, Docker, HTML, PHP, DevOps ...)
Hardware Design & MCUs
(e.g. ATmega128, MSP430, 80C51, TDA4)
Operational Research
(e.g. LP, MIP, CP ...)

Languages
Mandarin
English

CERTIFICATES

- JHU certified Data Science Specialization
- Bk certified Big Data Analysis with Spark
- GE certified Green Belt of Lean Six Sigma
- CN Automation certified Mid-Class Eng

WORK HISTORY

Chief AI Scientist & Sr. Mgr
08/2019 - Now
UAES, CN
Lead AI lab of 10+ experts, best employee of 2020 for AI+Manufacturing solution (win SAIC innovation award two times and reported in SH TV); delivered a corp-wise platform for AI modeling in 2021; AI+product like PK, Zone, VCP, BMS...(2021-25) and published 15 papers, 50+ patents; Pearl Engineer of Pudong; Delivered corp-wise bot and LLM agent (2024-25) monthly serving 15% staff

Senior Researcher
09/2018 - 08/2019
Tencent AI Lab, CN
Lead research on deep learning algorithm for medical pathological diagnosis; 2 SCIE papers, 1 top-rank conference

Staff Data Scientist
04/2017 - 09/2018
GE Digital, CN
Lead research and deliver deep annotation algorithm for DCAR product, predictive maintenance for large healthcare equipment; deliver web platform for radiomics study in hospital; 1 top-rank conference, 1 US patent

Lead Engineer
05/2012 - 04/2017
GE Global Research, CN
Lead research on offshore drilling ultrasonic velocimetry, lift solution optimization for well lifecycle management, detection algorithm & physical modeling for micro-calcifications twinkling study, automation tool for GE controllers; 1 SCIE paper, 1 top-rank conference, 1 CTO award, 3 US patents

Scientist
09/2010 - 05/2012
Philips Research, CN
Research on signal processing algorithm for ultrasound blood velocimetry, and denoising algorithm for motion artifacts on ECG signal; 2 US patents

Hardware Engineer
09/2003 - 01/2004
Baokang Electronic Control, CN

EDUCATION

PhD of Computer Science
05/2008 - 09/2011
University of Aizu, JP
Research on computational model & 3D visualization for cardiac electrophysiological study; 1 SCIE paper, 2 conferences

PhD of Electronic Engineering
09/2004 - 06/2009
Fudan University, CN
Research on epi-cardial mapping system, including its data acquisition hardware, firmware, USB driver, 3D interpolation algorithm; 1 SCIE paper, 1 Chinese top-rank journal paper, 3 conferences

BSc of Electronic Engineering
09/1999 - 07/2003
Nanjing University of Sc. and Tec., CN
Major in radar system and signal processing

ACHIEVEMENTS, HONOURS AND AWARDS

- CTO Physical & Digital Integration Award, GE, 2016
- Best Employee, UAES, 2020
- 1st prize of CMQMA Excellent Quality Management, CN, 2022
- Pearl Engineer, Pudong Shanghai, 2024

RECOMMENDATIONS

"...Weijia has developed an excellent reputation within our organization as a dedicated, insightful and easy to work with colleague..." - by Chief Engineer @ GE US Probes

FEATURED PROJECT AND BUSINESS IMPACT

AI + Manufacturing Projects
Computer Vision for PCBA Assisted Visual Inspection [24,35,36], 2022
Project Overview: Leveraged deep learning networks with contrastive learning to enhance model generalization. Post-processing operators were designed based on insights from veteran experts, ensuring strong interpretability across the entire solution.
Key Achievements: Deployed to mass production in June 2021; continues to serve all company PCBA production lines. Achieved zero missed detections with less than 10% false positive rate. Reduced manual visual inspection workload by over 92% at the Shanghai plant alone. Won two SAIC Innovation Awards and was featured in a report by Shanghai Television.

Smart Sensor Projects
Automatic annotation algorithm for arrhythmia [15], 2017-2018
Project Overview: My first AI project and publication designed for GE diagnostic cardiology (DCAR) portfolio. Multichannel signals are transferred into frequency spectrum. Then a latent representation are learned by VAE.
Key Achievements: Integrated into DCAR product.

AI + Contrl Projects
Battery Balancing Algorithm for BMS [32,33,38], 2023
Project Overview: Designed for passive balancing scenarios, the objective was to identify optimal strategies that enable the controller to guide the circuit—avoiding thermal protection triggers while minimizing balancing time. including:
- Built a mechanism model using Simscape
- Calibrated hyperparameters with physical measurement data
- Developed a customized NeuralODE to fit the mechanism model, creating an AI surrogate model
- Constructed a Gym environment
- Learned balancing strategies via reinforcement learning
- Analyzed RL-derived strategies and simplified them into robust heuristic strategies

Key Achievements: Integrated into BMS product in 2024, improving balancing efficiency by over 10% compared to conventional strategies.

AI + Supply Chain Projects
APAS Algorithm [31], 2020 & 2025
Project Overview: Utilized mixed-integer programming (MIP) and constraint programming (CP) from operations research. Given order demands, production line manufacturability constraints, and tooling cart constraints, the system determines optimal capacity planning and production scheduling.
Key Achievements: Implemented in two phases.
- Phase 1 (April 2020): Capacity planning module launched in the IE system, using the CBC solver.
- Phase 2 (2025): Production scheduling algorithm launched in September, using IBM CPLEX. Currently operational at the Liuzhou factory.

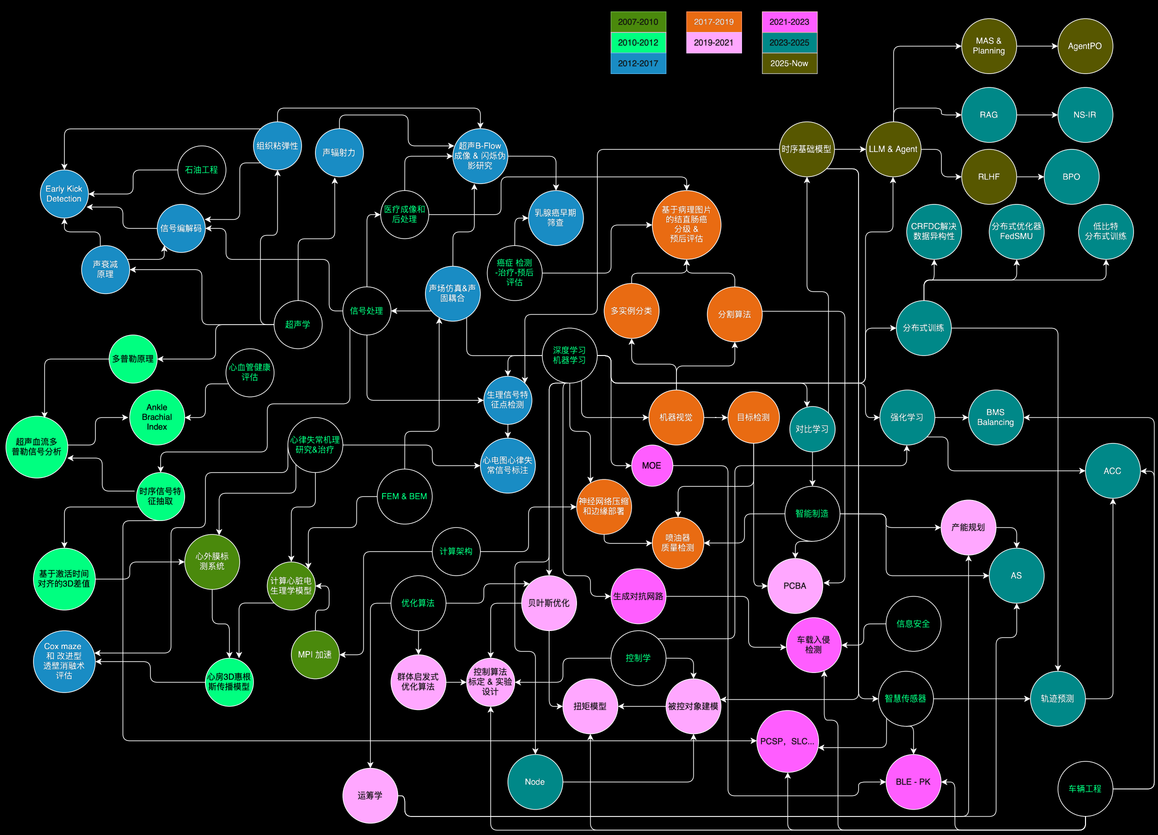
Algorithm vs. manual capacity planning: 5.2% labor hour savings, 2.7% equipment time savings.
Algorithm vs. manual production scheduling (at 100% capacity utilization): 6% reduction in order lead time, 1.8% fewer tooling cart changeovers.

Agents Projects
Agents for R&D Efficiency [37,46,47,48,49,50,52,53], 2023 –Present
Project Overview: By combining large language models, context engineering, ontology and knowledge curation, MAS reasoning, and tool orchestration, the initiative comprehensively empowers the company's R&D processes, including two major R&D portals.
Key Achievements: Improvements in the system development ASPICE process model: System Requirements Analysis (+5%), System Architecture Design (+11%), System Integration (+8%), ASW Software (+3%–9%), BSW Software (+2%–18%). Skill duplicate loading rate: 64%. Portal bots serve 15% of employees monthly. The self-developed PDF structured parsing tool averages 1,000+ daily uses. And copilot product called Ulystudio in USP 3.0.

RESEARCH OVERVIEW

30+ papers, 60+ patents, those with following symbols are recommended: (♣) Software implementation and algorithm; (♠) Signal Processing, deep/machine learning and computer architecture, AI for control, plant modeling; (♡) Physics and computational modeling; (♢) Optimization algorithm and operational research

Weijia's research can be broadly divided into the following stages:
2007–2010: Specialized in medical electrophysiology research, investigating arrhythmia detection and mechanisms through hardware system design, software development, time-series signal processing algorithms, and computer simulation. Simulations encompassed cell transmembrane potential models, Huygens propagation models, and electric field simulation using the boundary element method.
2010–2017: Expanded research to medical, oil & gas, and other sectors, applying signal processing and computer simulation to address industry-specific challenges. Signal processing evolved from 1D time-series signals to 2D ultrasound images, while simulations extended from electric fields to acoustic fields, fluid-structure coupling, acoustic radiation computation, medium attenuation, and ultrasound imaging.
2017–2019: With TensorFlow's emergence, AI was adopted for electrophysiological signal processing starting in 2016. Then focused on AI-driven analysis of medical pathology images, including instance segmentation, cancer grading based on segmentation results, and prognosis assessment from histopathological images.
2019–Present: Leading a team dedicated to solving critical automotive industry challenges with AI, thus provides direct product augmentation. e.g. computer vision for industrial manufacturing, optimization algorithms for accelerated calibration, AI+information security, smart sensor development, AI+control systems, plant modeling, and federated training.
2023–Present: With the advent of large models, leading team to build AI agents and company-wide ChatBots. Research contributions focus on information retrieval, large model inference, ontology, and knowledge engineering.
Here's a diagram briefly shows the developing of research topic (A interactive version please goto this link):



Moreover, if you need details, please follow Weijia's Wechat Official Account (ALF & SAM) and use AskMe Bot in service.

FEATURED PUBLICATIONS

1. Design and Implementation of a New System for Whole-Atrial Epicardial Mapping

Cuiwei Yang, Weijia Lu, Xiaomei Wu, and Zuxiang Fang

2007 International Journal of Bioelectromagnetism (IJB) Link

About: *This study introduces the way to design an electronic system to records electrophysiology activity of heart. Cardiac Mapping and Cardiac Modeling.*

2. A New Scheme for Observation and Interpretation Atrial Fibrillation

Weijia Lu, Zuxiang Fang

2008 in Proceedings of the 2nd International Conference on Bioinformatics and Biomedical Engineering (ICBBE) Link

3. A Visual Expression to Show Epicardial Electrical Activity Comprehensively

Tou Zhou, Weijia Lu, Cuiwei Yang, and Zuxiang Fang

2008 in Proceedings of the 2nd International Conference on Bioinformatics and Biomedical Engineering (ICBBE) Link

4. Dynamic Epicardial Mapping Using 3D Emulation

Weijia Lu, Tuo Zhou, Cuiwei Yang, and Zuxiang Fang

2008 in Proceedings of the International Conference on Biomedical Engineering and Informatics (ICBEI) Link

5. Development of Epicardial Mapping System for Study Atrial Fibrillation

Cuiwei Yang, Weijia Lu, Tuo Zhou, Xiaomei Wu, and Zuxiang Fang

2008 in Proceedings of the International Conference on Biomedical Engineering and Informatics (ICBEI) Link

6. A Method for Real-time Sampling and Smoothly Scrolling in Epicardial Mapping System (♣)

Weijia Lu, Cuiwei Yang, and Zuxiang Fang

2009 Journal of Biomedical Engineering (JBE Chinese), vol.26, pp.1102-1105 Link

About: *This study introduces software design of employing DirectX to smoothly scrolling multiple signals on screen in a high speed sampling scenario. The corresponding GUI of system reported in IJB2007. Cardiac Mapping and Cardiac Modeling.*

7. A Parallel Algorithm for Computer Simulation of Electrocardiogram Based on MPI (♣)

Wenfeng Shen, Weijia Lu, Daming Wei, Weimin Xu, Xin Zhu, and Shizhong Yuan

2009 in Proceedings of 8th IEEE/ACIS International Conference on Computer and Information Science (ICIS) Link

About: *This study is about software design of ECG computational simulation in HPC. A HPC version of Wei-Harumi Model. And the Wei-Harumi Model is a pioneering whole-heart computer simulation model for cardiac electrophysiology, based on Huygens' Principle Propagation and developed by Dr. Daming Wei. Cardiac Mapping and Cardiac Modeling.*

8. Implementation of a Novel Interpolating Method to Epicardial Potential Mapping for Atrial Fibrillation Study (♣♣♣)

Weijia Lu, Cuiwei Yang, Zuxiang Fang, Xingpeng Liu, Xin Zhu, and Daming Wei

2010 Computers in Biology and Medicine (CBM), vol.40, pp.456-463 Link

About: *An Spatio-Temporal interpolation algorithm to estimation a high resolution electrophysiology field. Traditional interpolation only consider the spatial relationship of sampling position while our method combines the spatial calculation with electrophysiology activity propagation. Designed for system reported in IJB2007. Cardiac Mapping and Cardiac Modeling.*

9. A Computer Model Based on Real Anatomy for Electrophysiology Study (♡♡♡♣)

Weijia Lu, Daming Wei, Xin Zhu, and Wenxi Chen

2011 Advances in Engineering Software (AES), vol.42, pp.463-476 Link

About: *An computational model builtup based on real anatomy information to simulate 12 channel ECG. A upgrading of Wei-Harumi Model by introducing real anatomy, cellular Ion Channel description and atrial propagation system. Especially for artrial arrhythmia study. And the Wei-Harumi Model is a pioneering whole-heart computer simulation model for cardiac electrophysiology, based on Huygens' Principle Propagation and developed by Dr. Daming Wei. Cardiac Mapping and Cardiac Modeling.*

10. Method and Device for Detecting Occlusion/Reopening of an Artery and System for Measuring Systolic Blood Pressure (♠)

Yinan Chen, Weijia Lu, Jianyi Zhong, Ajay Anand, John Petruzzello

2012 US 10076310 B2 Link

About: *Method to detecting blood pressure using pulse wave ultrasound. First achievement during my career path fulfilled in Philips Research. Handcrafted signal feature is extracted from IQ data to predict the state of artery. Medical Ultrasonic and Signal Processing.*

11. Computer Simulation of Cathode Ablation for Atrial Fibrillation (♡)

Xin Zhu, Di Yang, Weijia Lu, Wenxi Chen, Daming Wei, Koji Fukuda, and Hiroaki Shimokawa

2014 in Proceedings of 14th IEEE International Conference on Computer and Information Technology (CIT) Link

About: *Simulation atrial fibrillation using computational model reported in AES2011 and created by Dr. Weijia Lu. The simulation results in this study demonstrated that non-transmural maze III ablation procedure can achieve the same effect as the transmural ablation if the positions of ablation lines are reasonable. Thus this paper serves as another clinic evidence of the efficiency of computational model reported in AES2011. Cardiac Mapping and Cardiac Modeling.*

12. Method to Develop Coded Excitation for Velocimetry in Downhole Drilling (♠♠♡)

Weijia Lu, Ran Niu, Longtao Yuan, Xin Qu, Heng Wu, Jing Ye

2015 in Proceedings of 15th IEEE International Conference on Computer and Information Technology (CIT) Link

About: *Encoding-decoding algorithm for pulse wave ultrasound, which can significantly improving spatial resolution without jeopardizing signal penetration. By-product of my studying related to B-Flow (JMU2017) and designed for early kick detection project, and my first publication in GE Research. Details are as following: In this paper, the phase modulation coding/decoding technology is discussed for the application in the high attenuation medium based on the simulation and the experiment. For the decoding perspective, the matched filter and the inverse filter are compared. An extra processing is introduced right after the matched filtering to restrain the range* *sidelobes. Furthermore, in the latter part of this paper, the factors to cause a degeneration in signal-noise ratio are quantitatively evaluated by simulation beam pattern and Doppler signal. Based on the experiments and simulation in this paper, it's found that the signal amplitude get significant boosted (30dB) while the range resolution is compatible with the span of a single pulse. It's also proved that the frequency dependent phase distortion from attenuation will cause more degeneration than Doppler effect, but even this influence can be compensated by the equalization. Industry Ultrasonic and Signal Processing.*

13. Dominant Factor Analysis of B-flow Twinkling Sign with Phantom and Simulation Data (♠♡♡♡)

Weijia Lu, Bruno Haider

2017 Journal of Medical Ultrasonics (JMU), vol.44, pp.37-50 Link

About: *A mechanistic study of twinkly phenomenon showup in B-Flow ultrasonic imaging. A study established a multiphysics computational model to describe the acoustic coupling of ultrasonic field and solid granules, then designed phantom study to validate the mechanistic theory. The agreement of simulation and phantom study is found by a post-processing algorithm in also reported in this study. The break-through achievement is acknowledged by the chief engineer of medical ultrasound BU. Medical Ultrasonic and Signal Processing.*

14. Sensing Systems and Methods for Detecting Changes in Downhole Hydrocarbon and Gas Species (♠♡)

Weijia Lu, Yi Liao

2017 WO2018170838A1 Link

About: *A study of composition analysis using ultrasound attenuation also for early kick detection. Industry Ultrasonic and Signal Processing.*

15. Method to Annotate Arrhythmias by Deep Network (♠♠♠)

Weijia Lu, Jie Shuai, Shuyan Gu, Joel Xue

2018 in Proceedings of 18th IEEE International Conference on Computer and Information Technology (CIT) Link

About: *Automatic annotation of arrhythmias on ECG signal for GE Diagnostic Cardiology portfolio. My first publication about deep learning and AI, my leading collaboration of GE Healthcare China and GE Healthcare US. AI for Diagnostic Cardiology.*

16. New Boundary Constraint Loss to Facilitate Glands Segmentation (♠♠♠)

Weijia Lu, Jianhua Yao, Xiao Han, Haocheng Shen

2019 Journal of Medical Imaging and Health Informatics (JMIHI) Link

About: *Segmentation of glands on pathological image, learning method by a new loss function. My first publication of using AI on medical image and fulfilled in Tencent AI Lab. AI for Pathologic Image.*

17. An Attentive Pruning Method for Edge Computing (♠)

Yang Gao, Hao Gong, Weijia Lu, Chen Su, Zhang Ni and Qinghua Wang

2019 in Proceedings of 20th International Conference on Machine Learning and Computing (ICMLC) Link

About: *Method to prune object detection network, first publication of UAES AI Lab after my enrollment of UAES as chief AI scientist. Embedded AI.*

18. System and Method for Identifying Cardiac Arrhythmias With Deep Neural Networks (♠♠♠)

Weijia Lu, Shuyan Gu, Joel Xue, Jie Shuai, Hu Lifei

2020 US 10869610 B2 Link

About: *The corresponding patent of publication CIT2018, this patent is authored by Joel, the principle engineer of GE Diagnostic Cardiology, and the coauthor of my publication CIT2018. AI for Diagnostic Cardiology.*

19. Microsatellite Instability Prediction of Uterine Corpus Endometrial Carcinoma Based on HE Histology Whole-Slide Imaging (♠♠♠)

Tongxin Wang, Weijia Lu, Fan Yang, Li Liu, Zhong-Yi Dong, Weimin Tang, Jia Chang, Wenjing Huan, Kun Huang and Jianhua Yao

2020 in Proceedings of IEEE 17th International Symposium on Biomedical Imaging (ISBI) Link

About: *An new AI paradigm to predict MSI on pathological image. A multi-instance learning method by Dr. Weijia Lu and Tongxin, when Tongxin was working as an intern in Tencent AI Lab. AI for Pathologic Image.*

20. Development and interpretation of a pathomics-based model for the prediction of microsatellite instability in Colorectal Cancer (♠♠♠)

Cao Rui, Fan Yang, Si-Cong Ma, Li Liu, Yu Zhao, Yan Li, Dehua Wu, Tongxin Wang, Weijia Lu, Wei-Jing Cai, Hong-bo Zhu, Xue-Jun Guo, Yuwen Lu, Jun-jie Kuang, Wenjing Huan, Wei-min, Tang, Kun Huang, Junzhou Huang, Jianhua Yao and Zhong-Yi Dong

2020 Theranostics Link

About: *A collaboration research fulfilled by Tencent AI Lab and Nanfang hospital, and a systematic description of methodology reported in ISBI2020. ISBI2020 is a study fulfilled by Dr. Weijia Lu and Tongxin, and this paper is an extension organized by Dr. Fan Yang. More experiments are reported in this paper. AI for Pathologic Image.*

21. Processing Methods, Devices, Equipment and Storage Media for Vehicle Data (♠)

Peng Liu, Weijia Lu, Bingyang Li, Hao Gong, Jie Zhuang and Tao Song

2020 CN 112598133 B Link

About: *Optimization of sampling point selection for utilizing gaussian process regression in torque prediction. Comparing with ASCMO modeling method by ETAS Bosch, our method use only 70% of sampling point without any decresing in prediction performance. AI for Smart Sensor.*

22. Construction Method, Device and Storage Medium for Engine Exhaust System Temperature Model (♢)

Bingyang Li, Hao Gong, Weijia Lu, Peng Liu, Chunshan Ma, Yang Wang, Jianqiang Wang and Zhiwei Wang

2021 CN 113239533 A Link

About: *A swarm intelligence method to optimize super-parameter map of the control algorithm. AI for smart calibration.*

23. Dual Batch Size Training: An efficient MGD adaptive batch size method

Yuhang Du, Wenfeng Shen, Baohua Liu, Weijia Lu and Hao Gong

2021 in Proceedings of 2021 IEEE 33rd International Conference on Tools with Artificial Intelligence, ICTAI Link

24. Method, Device and Storage Medium of PCB Welding Defect Detection (♠♠)

Weijia Lu, Peng Liu, Bingyang Li, Chuang Liu, Wei Shen, Huan Ge, Yu Jing, Jie Zhang, Qi Wang and Yu Cao

2021 CN 113506243 B Link

About: *A two-stage method to predict welding defect areas and defect type based on selective welding image. Segmentation AI model as well as hardcrafted feature from segmentation result are introduced in this patent. AI for manufacturing.*

25. Method, Device and Storage Medium of Image Recognition for Chip Welding Defect (♠)

Peng Liu, Weijia Lu, Bingyang Li, Chuang Liu, Tong Ma and Fayu Qian

2021 CN 113724218 B Link

About: *Welding defect detection based on resistance welding image. AI for manufacturing.*

26. Using EBGAN for Anomaly Intrusion Detection (♠♠)

Yi Cui, Wenfeng Shen, Jian Zhang, Weijia Lu, Chuang Liu, Lingge Sun and Sisi Chen

2022 in Proceedings of 2022 International Joint Conference on Neural Networks, IJCNN Link

About: *A generative adversarial network to detect intrusion on vehicle gateway. AI for Vehicle Safety.*

27. Knock detection method and device for PCSP ignition strategy (♠)

Xiaofeng Ma, Weijia Lu, Gang Xi and Jianqiang Wang

2022 CN 114781425 B Link

About: *A new knock detection method when traditional algorithm failed on PCSP ignition strategy. This method deal with labeling noise by cross training and carefully designed hand-crafted features. AI for Smart Sensor.*

28. Gradient-Based Meta-Learning Using Uncertainty to Weigh Loss for Few-Shot Learning

Lin Ding, Wenfeng Shen, Weijia Lu, Peng Liu and Shengbo Chen

2023 in Proceedings of ICCECE Link

About: *Extended from Model-Agnostic Meta-Learning (MAML), we propose two improved methods for the meta-loss part: Method 1 generates weights by comparing meta-loss differences to improve the accuracy when there are few classes, and Method 2 introduces the homoscedastic uncertainty of each task to weigh multiple losses based on the original gradient descent,as a way to enhance the generalization ability to novel classes while ensuring accuracy improvement*

29. Towards Designing an Attentive Deep Trajectory Predictor Based on Bluetooth Low Energy Signal (♠♠♠)

Weijia Lu, Xiaofeng Ma, Xiaodong Zhang, Zhifei Yang and Qinghua Wang

2023 in Proceedings of 57th Annual Conference on Information Sciences and Systems (CISS) Link

About: *A small but carefully designed MOE network to predict cellphone location in a key-less entry scenario of vehicle. The deep learning network, with only 700 floating parameters, has been deployed in a ECU with 300MHz frequency and limited code segment. This network has two branches, one to predict the angle and another one for radial distance, and whole network is sparse activated. Moreover a carefully designed loss function is reported in this study to accelerate network training. AI for Smart Sensor.*

30. Distributed Training Methods and Systems for Models (♠♠)

Weijia Lu, Xiaodong Zhang, Zhifei Yang, Xiaofeng Ma, Chuang Liu and Wangchen Lin

2023 CN 116822619 B Link

31. A Method for Automatic Capacity Allocation (♢)

Shuyu Jiang, Weijia Lu, Na Li, Huan Ge and Bingyang Li

2023 CN 116384669 A Link

About: *Automatic production line allocation using linear programming and cbc solver. AI for Scheduling.*

32. A Power Battery Balancing Controller, Balancing Control Method, and Electric Vehicle (♠♠♠)

Chuang Liu, Xichun Ke, Zhifei Yang, Weijia Lu, Xiaodong Zhang and Xiang Di

2023 CN 116674432 A Link

About: *A heuristic strategy for battery balancing. In nowadays, heterogenization of power battery cells becomes a critical factor of e-car lifespan. Certain chip has been provided to automatically initiate balancing process and ultimately ameliorate the heterogenization. But the chip will shutdown balancing once the temperature reaches a pre-set threshold. So this patent introduce a control-policy to fulfilled balancing process without trigger the temperature protection strategy. AI for Control.*

33. A Reinforcement Learning-based Battery Balancing Method and Device (♠♠)

Zhifei Yang, Xichun Ke, Chuang Liu, Weijia Lu, Xiaodong Zhang and Xiang Di

2023 CN 116767024 A Link

About: *The reinformant learning verion of CN 116674432 A. Moreover this patent reports the method to establish the digital twin model of the balancing hardware. This digital model is used as the environment during policy training. AI for Control.*

34. A Curve Information Processing Method, Device, Storage Medium, and Detection Equipment (♠)

Peng Liu, Lin Sun, Weijia Lu and Tong Ma

2023 CN 115631139 A Link

About: *AI for Manufacturing.*

35. A Target Detection Method, Device, Storage Medium, Sensor, and Controller (♠♠♠)

Peng Liu, Weijia Lu, Lin Sun, Can Zhang and Tong Ma

2023 CN 116452916 A Link

About: *Contrastive learning method for target detection. AI for Manufacturing.*

36. A Target Detection Method, Machine Vision Device, Storage Medium, and Controller (♠♠♠)

Peng Liu, Lin Sun, Weijia Lu, Jie Zhang, Wei Shen, Yu Jin and Huan Ge

2024 CN 117726855 A Link

About: *Semi-Supervised learning method for target detection. AI for Manufacturing.*

37. A Product Testing Method, Data Management Method, Apparatus, Medium and Controller (♠♠)

Weijia Lu, Xiaodong Zhang, Can Zhang, Zhifei Yang, Xiaofeng Ma, Chuang Liu, Bingyang Li, Feng Wu, Xuzhou Zhang, Jing Ye, Yongyi Liu, Xichun Ke, Jianfei Zheng, Jie Bai and Chen Sheng

2024 CN 118860853 A Link

About: *A test case generation tool utilizing signal matrix, IO configuration and large language model. LLM for Software Development.*

38. A Method for Model Data Processing, a Simulation Apparatus, a Storage Medium, and a Testing System. (♠♠♠)

Zhifei Yang, Xiaofeng Ma, Weijia Lu, Xiaodong Zhang, Wangchen Lin, Ting Li, Fei Sun, Qiang Fang and Gang Xi

2024 CN 118732531 B Link

About: *A method to establish digital twin model based on a new neural ODE structure. The method reported in this patent is mainly for time-series problem. AI for Plant Modeling.*

39. Scenario-Aware Clustered Federated Learning for Vehicle Trajectory Prediction with Non-IID Data (♠)

Liang Tao, Yangguang Cui, Xiaodong Zhang, Wenfeng Shen, Weijia Lu

2024 Part D: Journal of Automobile Engineering (PartD) Link

About: *A vehicle trajectory model, federated learned from real vehicle data, with multi-head design and federated clustering. The corresponding method is protected in patent CN 116822619 A. Federated Learning in Vehicle.*

40. A Comfortable and Robust DRL-based Car-following Policy Incorporating Lateral Information under Cut-in Scenarios (♠♠♠)

Yifei Shen, Zhifei Yang, Weijia Lu, Wenfeng Shen, Zhou Lei

2024 in Proceedings of 35th IEEE Intelligent Vehicles Symposium (IV) Link

About: *A reinforcement learning policy to significant increase the safety. The vehicle trajectory model, federated learned from real vehicle data and reported in PartD 2024, provide critical lateral information. AI for Control.*

41. Improving Generalization and Personalization in Long-Tailed Federated Learning via Classifier Retraining (♠♠)

Yuhang Li, Liu Tong, Wenfeng Shen, Yangguang Cui, Weijia Lu

2024 in Proceedings of 30th International European Conference on Parallel and Distributed Computing (Euro-par) Link

About: *A resampling strategy to address heterogenization issue of data distribution in a federated learning. Federated Learning in Vehicle.*

42. Addressing Sim2Real Challenges in DRL-based Car-following Policies via Federated Reinforcement Learning and Augmented State Observations (♠)

Yifei Shen, Zhifei Yang, Weijia Lu, Wenfeng Shen, Xiaodong Zhang, Xiaofeng Ma

2024 Proceedings of the 2024 International Conference on Intelligent Driving and Smart Transportation Link

43. Hybrid Attention with Memory-based Conditional Refinement for Unified Intent Trajectory Prediction (♠♠♠)

Jingjing Wang, Weijia Lu, Wenfeng Shen, Yangguang Cui, Tong Liu, Xiaodong Zhang, Zhifei Yang, Guangtai Ding, Yifei Shen, Liang Tao

2025 Transportation Research Record (TRR) Link

About: *A spatiotemporal network with semantic-search based training method. To improve the prediction of driver intention. AI for smart sensor.*

44. FedSMU: Communication-Efficient and Generalization-Enhanced Federated Learning through Symbolic Model Updates (♠♠♠)

Xinyi Lu, Hao Zhang, Chenglin Li, Weijia Lu, ZHIFEI YANG, Wenrui Dai, xiaodong Zhang, Xiaofeng Ma, Can Zhang, Junni Zou, Hongkai Xiong

2025 International Conference on Machine Learning (ICML) '25 Link

About: *Alleviating data heterogeneity by symbolizing client model updates (normalizing parameter magnitude) and combining Lion optimizer's local-global operations reduces communication costs and enhances the global model's generalization. Federated Learning.*

45. LBI-FL: Low-Bit Integerized Federated Learning with Temporally Dynamic Bit-Width Allocation (♠♠♠)

Li Ding, Hao Zhang, Wenrui Dai, Chenglin Li, Weijia Lu, ZHIFEI YANG, xiaodong Zhang, Xiaofeng Ma, Junni Zou, Hongkai Xiong

2025 International Conference on Machine Learning (ICML) '25 Link

About: *By using low-bit quantization and dynamic bit-width allocation via reinforcement learning, communication and computation costs are significantly reduced while maintaining high precision. Federated Learning.*

46. Logical Consistency is Vital: Neural-Symbolic Information Retrieval for Negative-Constraint Queries (♠♠♠)

Ganlin Xu, Zhoujia Zhang, Wangyi Mei, Jiaqing Liang, Weijia Lu, xiaodong Zhang, ZHIFEI YANG, Xiaofeng Ma, Yanghua Xiao, Deqing Yang

2025 The 63rd Annual Meeting of the Association for Computational Linguistics (ACL) Link

About: *A neuro-symbolic info retrieval method using first-order logic to boost complex query retrieval. A simple but effective compensation for naive semantic-based RAG system. Information retrieval for LLM.*

47. BPO: Revisiting Preference Modeling in Direct Preference Optimization(♠♠)

Lin Sun, Chuang Liu, Peng Liu, Bingyang Li, Weijia Lu, Ning Wu

2025 arXiv Link

About: *Direct Preference Optimization (DPO) have emerged as a popular method for aligning Large Language Models (LLMs) with human preferences. While DPO effectively preserves the relative ordering between chosen and rejected responses through pairwise ranking losses, it often neglects absolute reward magnitudes. This oversight can decrease the likelihood of chosen responses and increase the risk of generating out-of-distribution responses, leading to poor performance. To address this* *issue, we propose Balanced Preference Optimization (BPO), a novel framework that dynamically balances the optimization of chosen and rejected responses through two key components: balanced reward margin and gap adaptor. Finetune of LLM.*

48. FreePRM: Training Process Reward Models Without Ground Truth Process Labels (♠)

Lin Sun, Chuang Liu, Xiaofeng Ma, Tao Yang, Weijia Lu, Ning Wu

2025 arXiv Link

About: *Recent advancements in Large Language Models (LLMs) have demonstrated that Process Reward Models (PRMs) play a crucial role in enhancing model performance. However, training PRMs typically requires step-level labels, either manually annotated or automatically generated, which can be costly and difficult to obtain at scale. To address this challenge, we introduce FreePRM, a weakly supervised framework for training PRMs without access to ground-truth step-level labels. Finetune of LLM.*

49. ComLQ: Benchmarking Complex Logical Queries in Information Retrieval (♠♠)

Ganlin Xu, Zhitao Yin, Linghao Zhang, Jiaqing Liang, Weijia Lu, xiaodong Zhang, ZHIFEI YANG, Sihang Jiang, Deqing Yang

2026 The Fortieth AAAI Conference on Artificial Intelligence, 2026 Link

About: *Information retrieval (IR) systems play a critical role in navigating information overload across various applications. We propose a novel method leveraging large language models (LLMs) to construct a new IR dataset, which comprises 2,909 queries and 11,251 candidate passages. Our experimental results under zero-shot settings demonstrate existing retrieval models' limited performance on complex logical queries, especially on queries with negation, exposing their inferior capabilities of modeling exclusion. Information retrieval for LLM.*

50. AgentPO: Enhancing Multi-Agent Collaboration via Reinforcement Learning (♠♠)

Lin Sun, Chuang Liu, Can Zhang, Yubin Wu, Weijia Lu, Ning Wu

2026 The Fourteenth International Conference on Learning Representations, 2026 Link

About: *Multi-Agent Systems (MAS) offer a powerful paradigm for solving complex problems through distributed reasoning and collaboration. However, their effectiveness is often hindered by the challenge of optimizing interactions among agents. To address this, we introduce AgentPO, a novel framework that directly optimizes agent collaboration. AgentPO employs reinforcement learning to train a specialized Collaborator agent, which refines its interaction policy to enhance overall system performance within a fixed multi-agent topology. We evaluated AgentPO on multiple mathematical reasoning tasks, where it consistently outperformed strong baselines. With Llama-3.2-3B-Instruct as the actor model, AgentPO achieves accuracy improvements of +1.8% and +7.2% over strong baselines like Role Assignment and EvoAgent, respectively. When using the larger Llama-3.1-8B-Instruct model, these gains increase to +5.6% and +11.3%. Crucially, AgentPO achieves these results with remarkable efficiency: it requires only 500 training samples and operates at just 7.8% of EvoAgent's inference cost, highlighting its superior scalability and practicality.*

51. ProSAR: Prototype-Guided Semantic Augmentation and Refinement for Time Series Contrastive Learning(♠♠♠)

Caiyi Yang, Chenglin Li, Hao Zhang, Weijia Lu, ZHIFEI YANG, Wenrui Dai, xiaodong Zhang, Xiaofeng Ma, Can Zhang, Junni Zou, Hongkai Xiong

2026 International Conference on Machine Learning (ICML) '26 Link

About: *The paper ingeniously adapts traditional temporal-frequency feature analysis methods for prototype matching and data augmentation pipelines, pairing them with learnable prototypes and contrastive learning to ultimately achieve state-of-the-art results on five multivariate time series forecasting datasets as well as the UEA time series classification benchmark.*